THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $1\frac{1}{2} \times 4$ equals: 1

(a) $4\frac{1}{2}$ **(b)** 3 **(c)** 6 **(d)** 12

2. Angles A and B are vertically opposite and $A = 28^\circ$. Then $B =$ 1

(a) 38° **(b)** 28° **(c)** 18° **(d)** 48°

3. The sum $1 + 2 + 3 + \cdots + 10$ is: 1

(a) 54 **(b)** 57 **(c)** 56 **(d)** 55

4. In $\triangle ABC$, the angle included between sides AB and AC is: 1

(a) $\angle A$ **(b)** $\angle C$ **(c)** $\angle B$ **(d)** $\angle A + \angle B$

5. The reciprocal of $1\frac{1}{2}$ (as a fraction) is: 1

(a) $\frac{11}{12}$ **(b)** $1\frac{1}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{2}{3}$

6. An angle of 35° is: 1

(a) reflex **(b)** acute **(c)** right **(d)** obtuse

7. A number is divisible by 5 if its last digit is: 1

(a) 0 or 2 **(b)** 5 only **(c)** any odd digit **(d)** 0 or 5

8. Which set of angles can be the angles of a triangle? 1

(a) $90^\circ, 90^\circ, 10^\circ$ **(b)** $50^\circ, 60^\circ, 70^\circ$ **(c)** $50^\circ, 60^\circ, 80^\circ$ **(d)** $100^\circ, 50^\circ, 40^\circ$

9. $\frac{1}{2} \div \frac{1}{4}$ equals: 1

(a) $\frac{1}{8}$ **(b)** 4 **(c)** 1 **(d)** 2

10. Three of the four angles around a point are 100°, 90° and 80°. The fourth angle is: 1

(a) 90° **(b)** 110° **(c)** 100° **(d)** 80°

11. The sum of five consecutive numbers with middle number 8 is: 1

(a) 40 **(b)** 39 **(c)** 41 **(d)** 42

12. In $\triangle ABC$, $AB = AC$. Then equal angles are: 1

(a) $\angle B = \angle C$ **(b)** all three **(c)** $\angle A = \angle C$ **(d)** $\angle A = \angle B$

13. A recipe uses $\frac{3}{4}$ cup of sugar for one batch. For 6 batches the sugar needed (cups) is: 1

(a) $4\frac{1}{2}$ **(b)** 9 **(c)** $2\frac{1}{4}$ **(d)** $4\frac{5}{6}$

14. If x° and 37° are complementary, then $x =$ 1

(a) 63° **(b)** 43° **(c)** 53° **(d)** 73°

15. The largest number that can be made using the digits $7, 5, 3, 1$ once each is: 1

(a) 1375 **(b)** 1357 **(c)** 7531 **(d)** 3715

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the digit $*$ so that $4 * 27$ is divisible by 9. 2

17. The angles of a triangle are $(2x + 10)^\circ$, $(3x - 4)^\circ$ and $(x + 12)^\circ$. Find x and the three angles. 2

18. Find $\frac{2}{3}$ of $\frac{3}{4}$ of 48 kg. 2

19. Find the angle that is 20° less than its supplement. 2

20. The sum of five consecutive even numbers is 110. Find the numbers. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Divide $2\frac{1}{2}$ by $\frac{3}{4}$ and write the answer as a mixed number. 3

22. Two complementary angles are in the ratio $1 : 2$. Find them. 3

23. Can 63 be written as the sum of three consecutive numbers? If yes, find them. 3

24. The exterior angle at C of $\triangle ABC$ is 120° and $\angle A : \angle B = 1 : 3$. Find all three angles of the triangle. 3

OR

One shirt needs $1\frac{1}{4}$ m of cloth. How much cloth is needed for 8 shirts? How many shirts can be made from 10\frac{5}{8} m? 3

25. Three of the four angles around a point are 120°, 70° and 60°. Find the fourth angle and name the kind of each of the four angles. 3

OR

Apply Kaprekar's routine twice to 2751. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. An angle is 2 times its supplement. Find the angle and its supplement. 5

OR

Write the Collatz sequence from 17. Count how many times the number is halved and how many times it is tripled and increased by 1. 5

27. Two sides of a triangle are 7 cm and 12 cm. 5

(i) Between which two lengths does the third side lie?

(ii) How many whole-number lengths are possible?

(iii) Find the smallest and the largest possible perimeter in whole centimetres.

OR

A garden is $\frac{4}{5}$ hm long and $\frac{5}{6}$ hm wide. $\frac{3}{4}$ of it is used for vegetables and the rest for flowers. Find the area of the garden, of the vegetable part and of the flower part (in hm^2). 5

END OF QUESTION PAPER Total: 50 Marks